



Habib University

Digital Logic and Design (EE-172 / CS-130 / CE-222)

Spring 2026

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Assignment No: 4

Release Date: 7th April, 2026

Due by: 24th April, 2026 11:59 PM

Total points: 100

Purpose:

The purpose of this assignment is to help you practice the number representation; its conversion; and the arithmetic operations in different base formats. In addition, this assignment will aid you to understand the process of representing a signed number in different formats; parity bits; representing complements of a number; and performing BCD arithmetic.

Instructions:

This assignment should be done individually.

All questions should be answered in **black ink only**.

Scan your answer sheet and upload it on LMS before the due date.

Grading Criteria:

Your assignments will be checked by instructor/TA.

You can also be asked to give a viva where you will be judged whether you understood the question yourself or not. If you are unable to correctly answer the question you have attempted right, you may lose your marks. No excuses/justifications will be entertained.

Zero will be given if the assignment is found to be plagiarized.

Untidy work or improper submissions will result in a reduction of your points.

Late submission penalty:

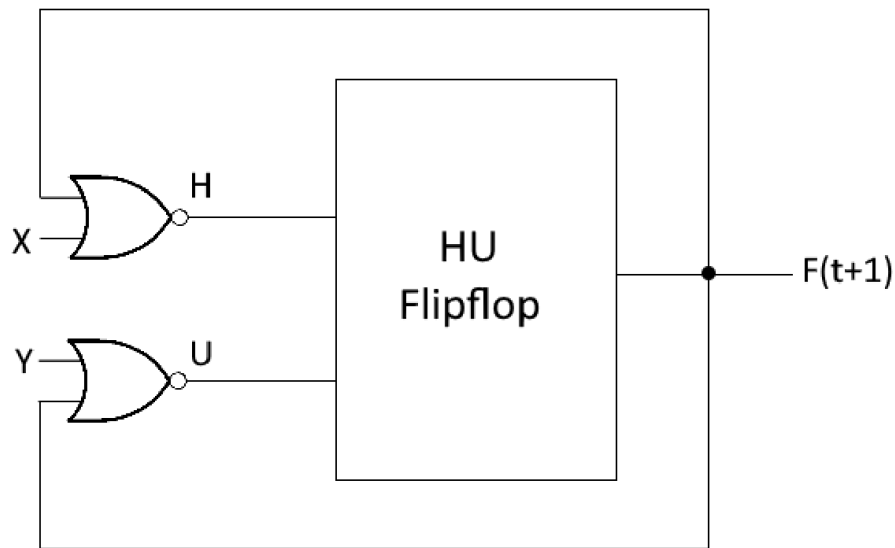
1-day late submission – 20% deduction of the maximum allowable marks

2-days late submission – 40% deduction of the maximum allowable marks

No submission will be accepted after two days of the original deadline

1. Consider the following sequential circuit:

3×4 + 4



The circuit has two inputs X and Y, and one state variable F. H and U represent the outputs at indicated points in the circuit diagram. The circuit requires a hypothetical flip flop that computes the next state of F using H and U in the following manner:

- i. No change when $HU = 00$
- ii. Set output to high when $HU = 11$
- iii. Complement current output when $HU = 01$, and
- iv. Clear output to 0 when $HU = 10$

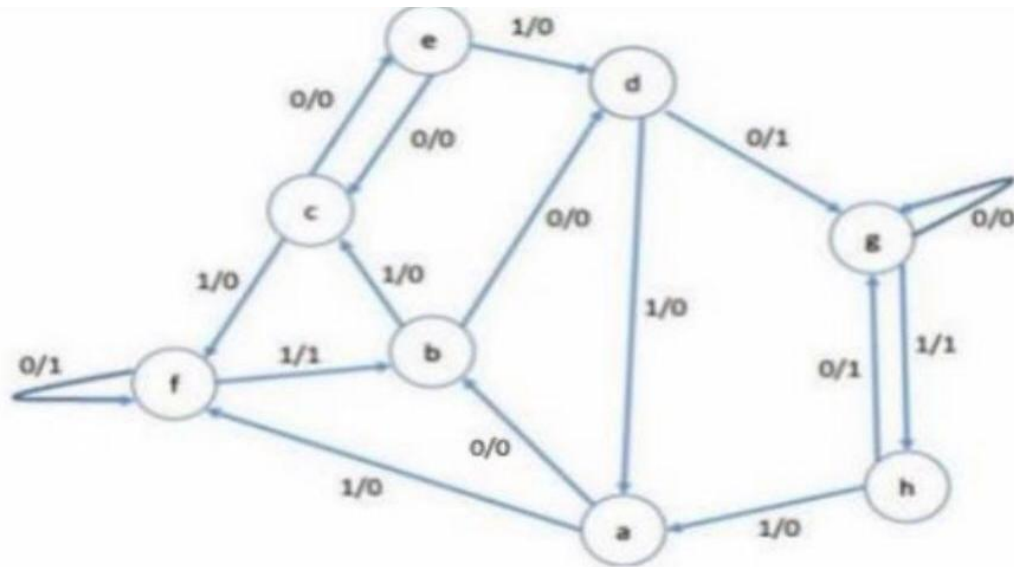
You can right away identify there is no such flip flop in the world, so you decide to define one of your own!

- a) Draw the characteristic table for your new flip flop HU.
- b) Derive the characteristic equation.
- c) Tabulate the excitation table.
- d) Obtain the state transition table for the given circuit.

For the given state diagram below:

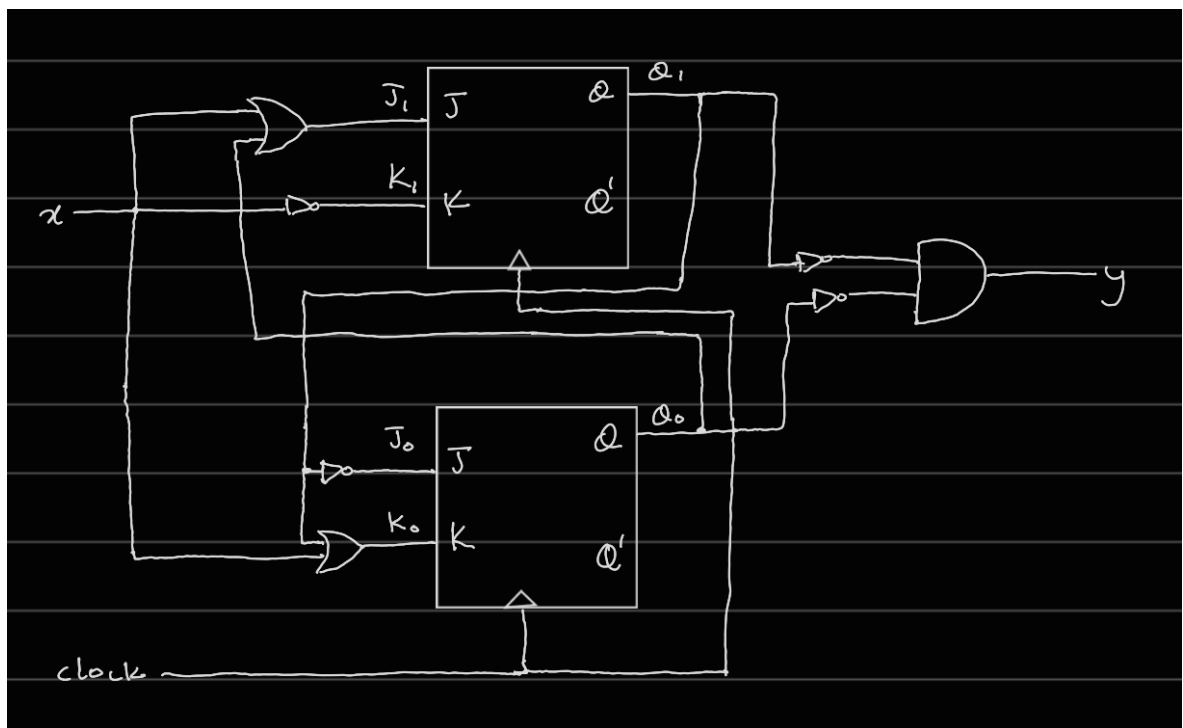
3+4+4+4

2.



- Draw the respective state table.
- Reduce the state table by identifying equivalent states using row comparison.
- Reduce further using the Implication Chart method.
- Draw the minimized state diagram.

3.



5 + 5 + 6

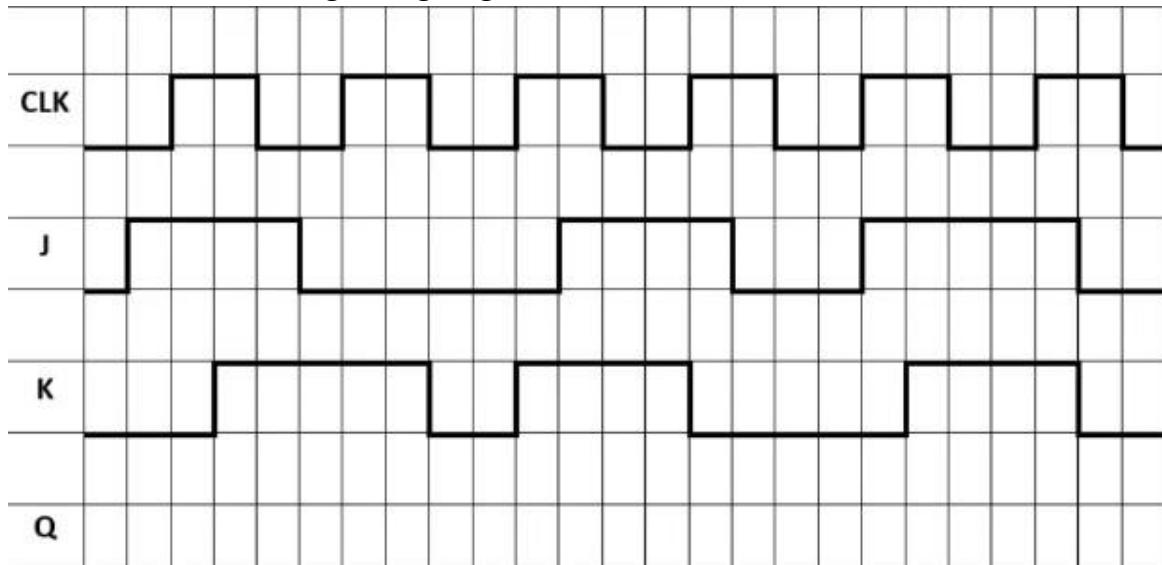
For the sequential circuit shown above, the circuit has one input x , two positive edge-triggered JK flip-flops with outputs Q_1 and Q_0 , and one output y . The flip-flop excitation inputs and output are defined as:

$$J_1 = x + Q_0 \quad K_1 = x' \quad J_0 = Q_1' \quad K_0 = Q_1 + x \quad y = Q_1'Q_0'$$

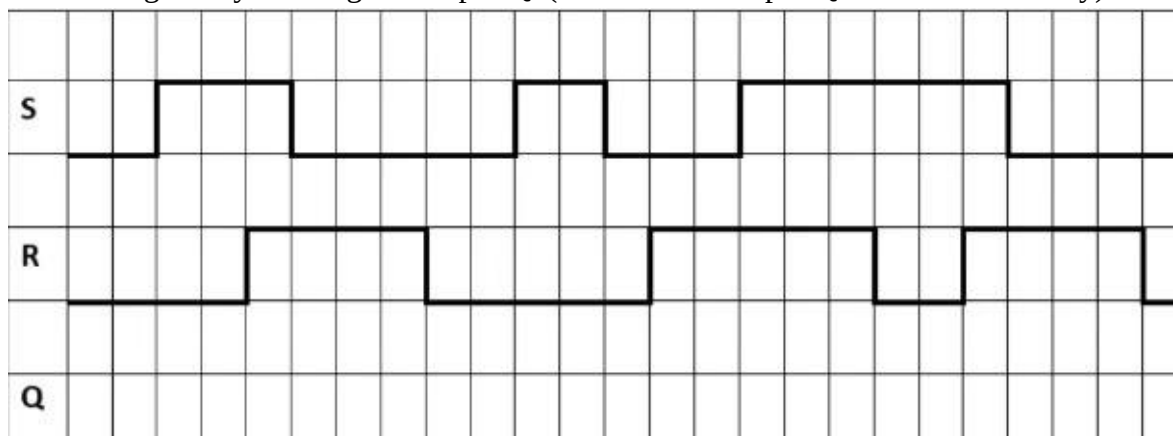
- Derive the next-state transition equations $Q_1(t+1)$ and $Q_0(t+1)$ using the JK characteristic equation and simplify fully.
- Create the state table defining the current states (Q_1, Q_0), input x , next states (Q_1^+, Q_0^+), all JK flip-flop excitation inputs (J_1, K_1, J_0, K_0), and output y .
- Draw the state diagram for the circuit. Specify whether this is a **Mealy or Moore** machine and justify your answer.

4.

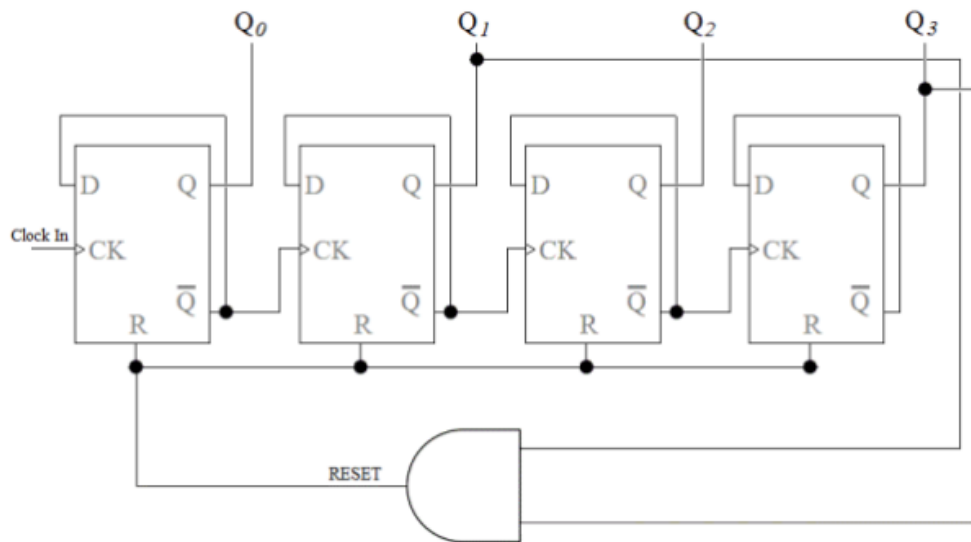
- a. Given the timing diagram of a positive edge triggered JK flip-flop, draw the output Q in the following timing diagram.



- b. The inputs for an SR-Latch are given in the following diagram. Complete the diagram by drawing the output Q. (Assume that output Q would be 0 initially)



- c. For the given asynchronous circuit,



Draw its timing diagram and determine the function of the circuit.

5	Design a sequential circuit using T flip-flops that simulates a digital clock module displaying the current time in 24 hour format. For example, if the current time is 23:12 then your circuit will display the binary value of 2, then 3, then 1 and then 2. - Draw the state diagram for the circuit above. - Draw the transition table for the circuit using T flip flops. - Use K-maps to deduce the equations for each T input for the flip flops. - Draw the circuit diagram for the sequential circuit.	4+4+2x6
6	You are designing a smart door security system that unlocks only when a specific sequence of sensor signals is detected. Design a digital lock for the door using JK flip-flop which opens if a person inputs the sequence “XYX”. You can consider that you only have two inputs ‘X’ and ‘Y’ and one output Z (locked/ unlocked). Each element of the sequence must last for one or more cycles, and there may be zero or more cycles of both inputs low between the sequence elements. After the lock is open, either input going high causes the unlock to go low. Draw the circuit diagram for a digital lock implementation and show all the relevant working (state table, state diagram, K-maps, state equations). Specify what kind of state machine this is (mealy or moore)	15